

Lab 3: Component Modelling & Architectural Pattern Selection

Smart Lab Equipment & Slot Reservation Portal

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Architecture Selection

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal System.

Reason 1: Clear Separation Between Interaction, Business Rules, and Records

The system has three distinct concerns: how students and Lab Technicians interact with the portal (Presentation Layer), how reservation, calibration, return, and restriction rules are enforced (Business Layer), and how equipment, reservation, and user records are persisted (Data Layer). Requirements such as FR-001 (slot reservation), FR-002 (calibration gating), FR-004 (late-return flagging), and FR-005 (disciplinary restriction) are all business rules that must be evaluated consistently regardless of which client (kiosk touch screen or web browser) submits the request. A layered structure keeps each rule in one dedicated component — Reservation Manager, Authentication & Access Control, Equipment & Calibration Manager, and Return & Disciplinary Management — so a change to one rule (e.g., extending the maximum booking window) never touches the UI or the database schema.

Reason 2: Predictable, Centralized Data Access Under Concurrency

NFR-001 requires concurrent slot-reservation requests to be processed within 200 ms with strict database-level lock isolation to prevent double-booking. In a layered design, every component that needs equipment, reservation, or restriction data goes through a single, well-defined Database component and its data interfaces (Reservation Data Interface, Equipment Data Interface, User Data Interface, Return & Restriction Data Interface). This gives one place to implement locking and transaction isolation, rather than scattering conflicting access logic across independently deployed services, which would be harder to reason about for a small-scale lab system.

Security Advantage

NFR-002 restricts equipment-management actions — updating calibration records and logging equipment returns — to authenticated users holding the Lab Technician role. The Authentication & Access Control component provides authentication and role verification before protected business operations are permitted. Keeping this component separate from the Reservation Manager, Equipment & Calibration Manager, and Return & Disciplinary Management components means role checks are implemented and maintained in one place instead of being duplicated across every business component that needs to enforce them.

Performance Benefit

Isolating persistence behind the Database component's Reservation Data Interface lets the team apply database-level lock isolation exactly where it is needed, without adding the network hops and serialization overhead that a microservices split would introduce. This keeps the reservation-locking path short and predictable, supporting the 200 ms response target in NFR-001 even under peak concurrent booking load.

Component Overview

- Kiosk / Web UI Component (Presentation) — handles student and technician interactions.
- Reservation Manager Component (Business) — slot search, booking, cancellation, concurrency control (FR-001, FR-003, NFR-001); checks calibration status and disciplinary restrictions before confirming a booking.
- Authentication & Access Control Component (Business) — login and role verification for students and Lab Technicians (NFR-002).
- Equipment & Calibration Manager Component (Business) — equipment catalog and calibration status (FR-002).
- Return & Disciplinary Management Component (Business) — logs equipment returns, flags late returns, and records/enforces disciplinary restrictions (FR-004, FR-005).
- Database Component (Data) — stores equipment, reservation, user, calibration, return, and restriction records.